Database Systems - Normalization. Normalization organizes the columns and tables of a relational database to reduce redundancy and prevent update anomalies. First normal form requires that every column holds a single atomic value and that each row can be identified by a primary key.

Second normal form removes partial dependencies, which means that every non key attribute must depend on the whole primary key and not on only a part of a composite key. Third normal form removes transitive dependencies, so a non key attribute must not depend on another non key attribute.

A transaction groups several database operations into one logical unit. The ACID properties describe atomicity, consistency, isolation and durability. Isolation levels such as read committed and serializable control which concurrent anomalies the database allows.